

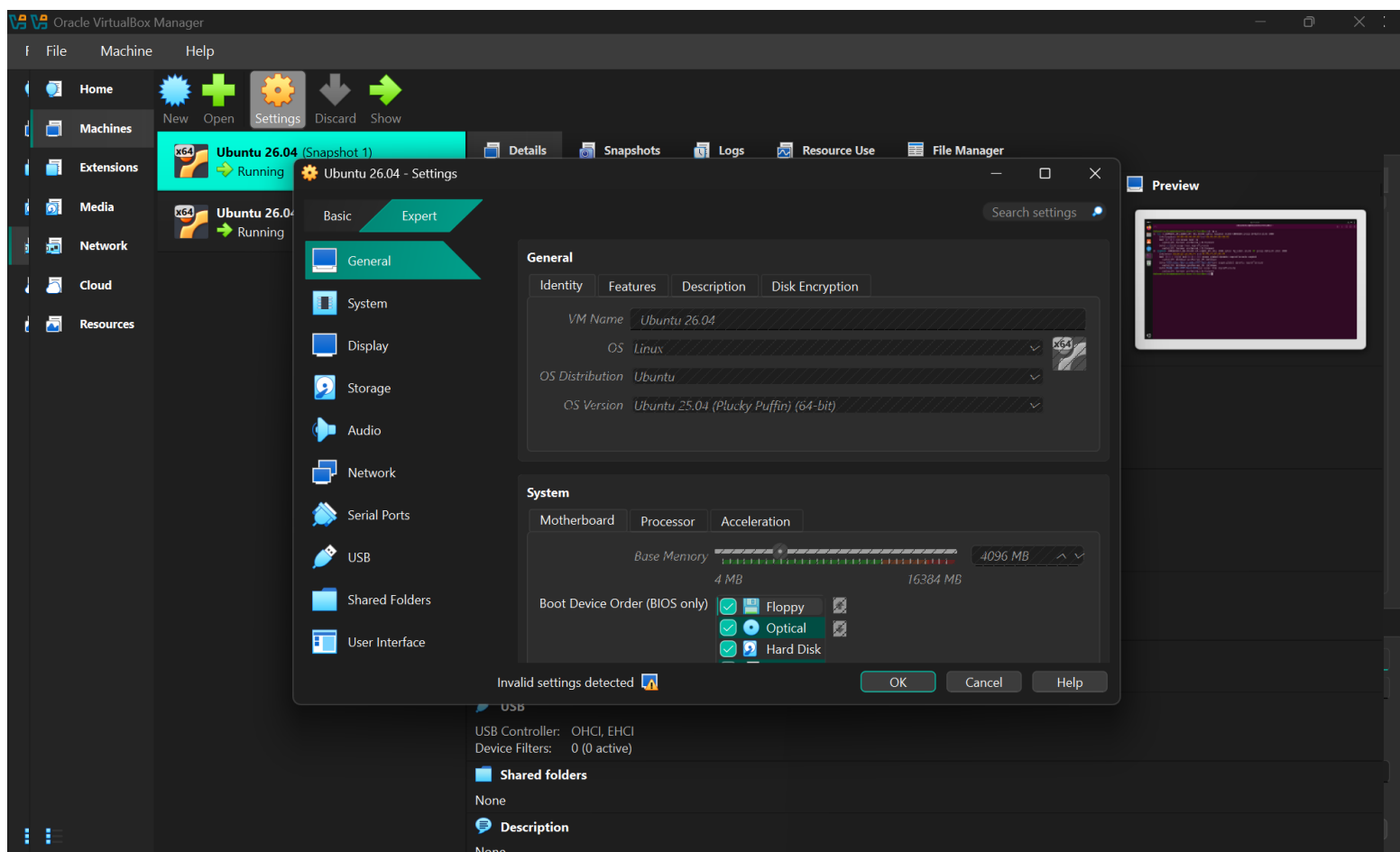
Experiment 2 — VM Cloning and Network Connectivity

Caption:

Figure 1: Creation and configuration of a customized NAT Network in Oracle VirtualBox

Details to write below the screenshot:

A customized NAT Network named **MyNatNetwork** was created in Oracle VirtualBox. The IPv4 prefix was configured as **10.0.2.0/24** and DHCP was enabled to assign IP addresses automatically to the connected virtual machines.



Screenshot 2 — Original VM Settings

Caption:

Figure 2: Network configuration of the original Ubuntu virtual machine

Details:

The network adapter of the original Ubuntu VM was configured to use the customized **NAT Network**. The network name was selected as **MyNatNetwork**, and the **Virtual Cable Connected** option was enabled.

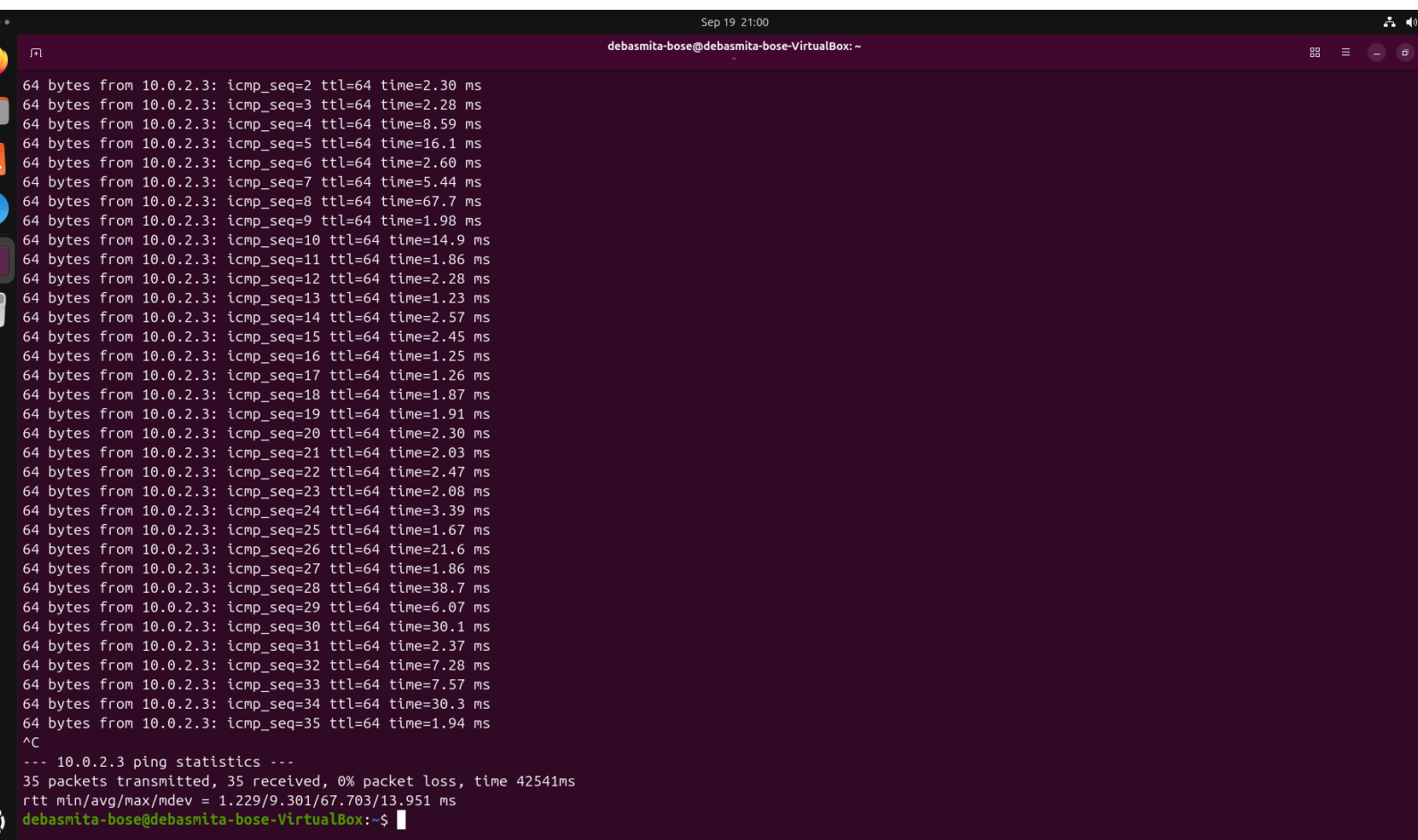
Screenshot 3 — Clone VM Settings

Caption:

Figure 3: Network configuration of the cloned Ubuntu virtual machine

Details:

The cloned Ubuntu VM was configured with the same **MyNatNetwork**. The network adapter was attached to the NAT Network and the virtual cable was enabled to establish network connectivity.



```
Sep 19 21:00
debasmita-bose@debasmita-bose-VirtualBox: ~
64 bytes from 10.0.2.3: icmp_seq=2 ttl=64 time=2.30 ms
64 bytes from 10.0.2.3: icmp_seq=3 ttl=64 time=2.28 ms
64 bytes from 10.0.2.3: icmp_seq=4 ttl=64 time=8.59 ms
64 bytes from 10.0.2.3: icmp_seq=5 ttl=64 time=16.1 ms
64 bytes from 10.0.2.3: icmp_seq=6 ttl=64 time=2.60 ms
64 bytes from 10.0.2.3: icmp_seq=7 ttl=64 time=5.44 ms
64 bytes from 10.0.2.3: icmp_seq=8 ttl=64 time=67.7 ms
64 bytes from 10.0.2.3: icmp_seq=9 ttl=64 time=1.98 ms
64 bytes from 10.0.2.3: icmp_seq=10 ttl=64 time=14.9 ms
64 bytes from 10.0.2.3: icmp_seq=11 ttl=64 time=1.86 ms
64 bytes from 10.0.2.3: icmp_seq=12 ttl=64 time=2.28 ms
64 bytes from 10.0.2.3: icmp_seq=13 ttl=64 time=1.23 ms
64 bytes from 10.0.2.3: icmp_seq=14 ttl=64 time=2.57 ms
64 bytes from 10.0.2.3: icmp_seq=15 ttl=64 time=2.45 ms
64 bytes from 10.0.2.3: icmp_seq=16 ttl=64 time=1.25 ms
64 bytes from 10.0.2.3: icmp_seq=17 ttl=64 time=1.26 ms
64 bytes from 10.0.2.3: icmp_seq=18 ttl=64 time=1.87 ms
64 bytes from 10.0.2.3: icmp_seq=19 ttl=64 time=1.91 ms
64 bytes from 10.0.2.3: icmp_seq=20 ttl=64 time=2.30 ms
64 bytes from 10.0.2.3: icmp_seq=21 ttl=64 time=2.03 ms
64 bytes from 10.0.2.3: icmp_seq=22 ttl=64 time=2.47 ms
64 bytes from 10.0.2.3: icmp_seq=23 ttl=64 time=2.08 ms
64 bytes from 10.0.2.3: icmp_seq=24 ttl=64 time=3.39 ms
64 bytes from 10.0.2.3: icmp_seq=25 ttl=64 time=1.67 ms
64 bytes from 10.0.2.3: icmp_seq=26 ttl=64 time=21.6 ms
64 bytes from 10.0.2.3: icmp_seq=27 ttl=64 time=1.86 ms
64 bytes from 10.0.2.3: icmp_seq=28 ttl=64 time=38.7 ms
64 bytes from 10.0.2.3: icmp_seq=29 ttl=64 time=6.07 ms
64 bytes from 10.0.2.3: icmp_seq=30 ttl=64 time=30.1 ms
64 bytes from 10.0.2.3: icmp_seq=31 ttl=64 time=2.37 ms
64 bytes from 10.0.2.3: icmp_seq=32 ttl=64 time=7.28 ms
64 bytes from 10.0.2.3: icmp_seq=33 ttl=64 time=7.57 ms
64 bytes from 10.0.2.3: icmp_seq=34 ttl=64 time=30.3 ms
64 bytes from 10.0.2.3: icmp_seq=35 ttl=64 time=1.94 ms
^C
--- 10.0.2.3 ping statistics ---
35 packets transmitted, 35 received, 0% packet loss, time 42541ms
rtt min/avg/max/mdev = 1.229/9.301/67.703/13.951 ms
debasmita-bose@debasmita-bose-VirtualBox:~$
```

Screenshot 4 — IP Configuration

Caption:

Figure 4: Verification of IP configuration using the ip a command

Details:

The ip a command was used to display the network interfaces and IP configuration of the VM. The enp0s3 interface was active and obtained the IP address **10.0.2.3/24**.

Screenshot 5 — Ping Test to VM-2

Caption:

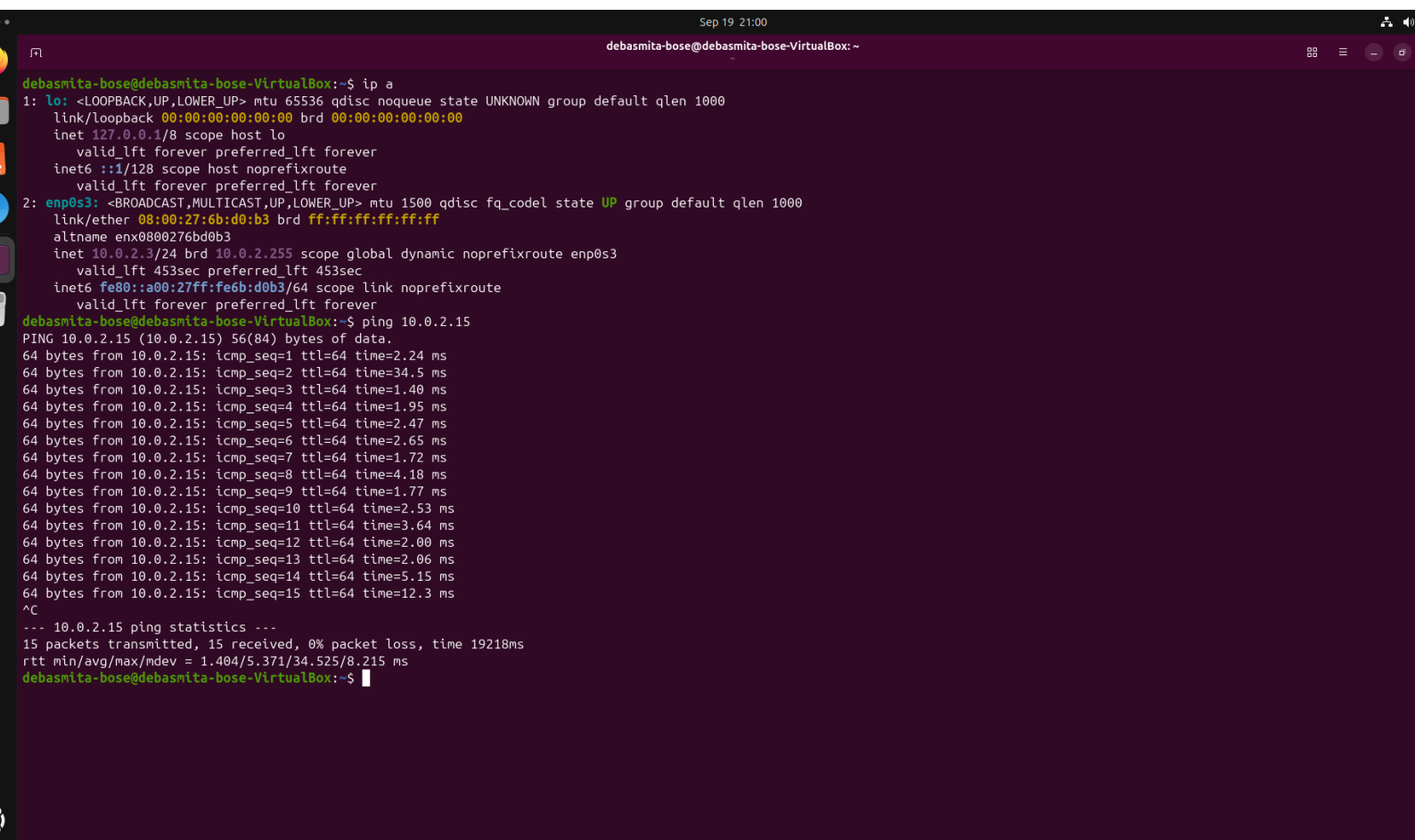
Figure 5: Successful connectivity test from VM-1 to VM-2

Command:

ping 10.0.2.15

Details:

The connectivity between the two virtual machines was tested using the ping command. Successful replies were received from **10.0.2.15**, with **0% packet loss**, confirming successful communication between the VMs.

A screenshot of a terminal window titled 'debasmita-bose@debasmita-bose-VirtualBox: ~'. The terminal shows the output of the 'ip a' command, displaying details for the loopback interface 'lo' and the ethernet interface 'enp0s3'. Below this, the command 'ping 10.0.2.15' is executed, resulting in 15 successful ping requests with 0% packet loss. The terminal output for the ping command is as follows:

```
debasmita-bose@debasmita-bose-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:6b:d0:b3 brd ff:ff:ff:ff:ff:ff
    altname enx0800276bd0b3
    inet 10.0.2.3/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 453sec preferred_lft 453sec
    inet6 fe80::a00:27ff:fe6b:d0b3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
debasmita-bose@debasmita-bose-VirtualBox:~$ ping 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data:
 64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=2.24 ms
 64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=34.5 ms
 64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=1.40 ms
 64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=1.95 ms
 64 bytes from 10.0.2.15: icmp_seq=5 ttl=64 time=2.47 ms
 64 bytes from 10.0.2.15: icmp_seq=6 ttl=64 time=2.65 ms
 64 bytes from 10.0.2.15: icmp_seq=7 ttl=64 time=1.72 ms
 64 bytes from 10.0.2.15: icmp_seq=8 ttl=64 time=4.18 ms
 64 bytes from 10.0.2.15: icmp_seq=9 ttl=64 time=1.77 ms
 64 bytes from 10.0.2.15: icmp_seq=10 ttl=64 time=2.53 ms
 64 bytes from 10.0.2.15: icmp_seq=11 ttl=64 time=3.64 ms
 64 bytes from 10.0.2.15: icmp_seq=12 ttl=64 time=2.00 ms
 64 bytes from 10.0.2.15: icmp_seq=13 ttl=64 time=2.06 ms
 64 bytes from 10.0.2.15: icmp_seq=14 ttl=64 time=5.15 ms
 64 bytes from 10.0.2.15: icmp_seq=15 ttl=64 time=12.3 ms
^C
--- 10.0.2.15 ping statistics ---
15 packets transmitted, 15 received, 0% packet loss, time 19218ms
rtt min/avg/max/mdev = 1.404/5.371/34.525/8.215 ms
debasmita-bose@debasmita-bose-VirtualBox:~$
```

Screenshot 6 — Reverse Ping Test

Caption:

Figure 6: Successful connectivity test from VM-2 to VM-1

Command:

ping 10.0.2.3

Details:

A reverse connectivity test was performed by pinging the IP address of the other VM. The VM received successful replies from **10.0.2.3**, with **0% packet loss**, confirming two-way communication.

Caption:

Figure 7: MAC address configuration of the virtual network adapters

Details:

The network adapter settings show the MAC address assigned to each virtual machine. The original and cloned VMs use separate virtual network adapters while being connected to the same NAT Network.